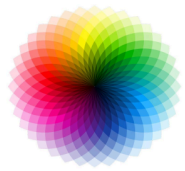


Dynamical Systems Meets Data:

DMD, Koopman, and Reduced Representations

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Dynamical Systems

A **dynamical system** describes how a state $\mathbf{x}(t)$ evolves over time t according to a fixed rule.

Discrete-time: $\mathbf{x}(t + \tau) = f(\mathbf{x}(t))$ Continuous-time: $\frac{d\mathbf{x}}{dt} = g(\mathbf{x}(t))$

- $\mathbf{x}(t) \in \mathbb{R}^n$: the state at time t
- f, g : a (possibly nonlinear) update rule
- Goal: Decompose the system into state space and temporal patterns, thus reducing temporal and state space dimensions.

Dynamical Systems (continued)

Linear Systems

If $f(\mathbf{x}) = \mathbf{A}\mathbf{x}$, the system is linear:

$$\mathbf{x}(t + \tau) = \mathbf{A}\mathbf{x}(t)$$

$\Rightarrow$ Solutions evolve through powers of $\mathbf{A}$: eigenvalues/eigenvectors govern behavior.

Dynamic Mode Decomposition

Assume that the data is sampled from the timeseries:

$$\mathbf{x}(t + \tau) \approx \mathbf{A} \mathbf{x}(t)$$

Decompose the system into spatial and temporal patterns.

Analyzing $\mathbf{A}$ results in the DMD (\citep{schmid2010dynamic}):

$$\mathbf{x}(t) = \sum_{j=1}^k \phi_j e^{\omega_j t} b_j$$

- $\phi_j \in \mathbb{C}^n$ **feature patterns** (dynamic modes)
- $\omega_j \in \mathbb{C}$ **temporal characteristics** (continuous time eigenvalues)
- $b_j \in \mathbb{R}$ scalar loadings (a.k.a. amplitudes)

Dynamic Mode Decomposition (math steps)

Starting from

$$\mathbf{x}(t + \tau) \approx \mathbf{A}\mathbf{x}(t)$$

iterate in discrete time ($m = t/\tau$):

$$\mathbf{x}(t) \approx \mathbf{A}^m \mathbf{x}(0)$$

If $\mathbf{A}\mathbf{W} = \mathbf{W}\mathbf{\Lambda}$ and $\mathbf{x}(0) = \mathbf{W}\mathbf{b}$, then

$$\mathbf{x}(t) \approx \mathbf{W}\mathbf{\Lambda}^m \mathbf{b} = \sum_{j=1}^k \phi_j \lambda_j^{t/\tau} b_j = \sum_{j=1}^k \phi_j e^{\omega_j t} b_j$$

Eigenvalue Interpretation

$$\lambda_j = e^{\omega_j \tau}$$

- Discrete time: λ_j
- Continuous time: ω_j

Useful conversion:

$$\omega_j = \frac{1}{\tau} \log(\lambda_j)$$

Eigenvalue Interpretation (polar form)

Write in polar form:

$$\lambda_j = r e^{i\theta}$$

- Trend: $r = |\lambda_j|$
- Oscillation frequency (per τ): $\theta = -i \log(\lambda_j/r)$

and with $\lambda_j = r e^{i\theta}$:

$$\omega_j = \frac{\log r + i\theta}{\tau}$$

Exact Dynamic Mode Decomposition (Dawson et al.)

1. Stack the data

$$\mathbf{X} = [\mathbf{x}(1) \mid \cdots \mid \mathbf{x}(T - \tau)] \in \mathbb{R}^{n \times p}, \quad \mathbf{X}' = [\mathbf{x}(1 + \tau) \mid \cdots \mid \mathbf{x}(T)] \in \mathbb{R}^{n \times p}$$

2. Want to solve

$$\min_{\mathbf{A}} \|\mathbf{X}' - \mathbf{A}\mathbf{X}\|_F$$

3. Rank- r truncated SVD $\mathbf{X} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ leads to

$$\begin{aligned} \mathbf{A}_\star &= \mathbf{X}'\mathbf{X}^\dagger, \quad \mathbf{X}^\dagger \approx \mathbf{V}\mathbf{\Sigma}^{-1}\mathbf{U}^\top \\ \tilde{\mathbf{A}} &= \mathbf{U}^\top \mathbf{X}'\mathbf{V}\mathbf{\Sigma}^{-1} \in \mathbb{R}^{k \times k}, \quad \tilde{\mathbf{A}}\mathbf{W} = \mathbf{W}\mathbf{\Lambda} \end{aligned}$$

Exact Dynamic Mode Decomposition (continued)

4. Dynamic modes

$$\phi_j = \frac{1}{\lambda_j} \mathbf{X}' \mathbf{V} \mathbf{\Sigma}^{-1} \mathbf{w}_j$$

5. Eigenvalues (discrete time)

$$\lambda_j = e^{\omega_j \tau}$$

6. Loadings found by solving

$$\Phi \mathbf{b} = \mathbf{x}(1)$$

Compact reconstruction form:

$$\mathbf{x}(t) \approx \Phi \operatorname{diag}(e^{\omega t}) \mathbf{b}$$

My Interpretation (easier?)

The system evolves in the low-dimensional reduced space mapped to by $\mathbf{U}^\top \in \mathbb{R}^{k \times n}$:

$$\mathbf{U}^\top \mathbf{x}(t + \tau) = \mathbf{z}(t + \tau) = \tilde{\mathbf{A}} \mathbf{z}(t) = \tilde{\mathbf{A}} \mathbf{U}^\top \mathbf{x}(t)$$

Eigen-decompose $\mathbf{A}$ into eigenvectors $\mathbf{w}_j$ and eigenvalues λ_j .

Projected dynamic modes:

$$\phi_j = \mathbf{U} \mathbf{w}_j$$

DMD analyzes the stability of this system and generates spatial patterns in $\mathbb{R}^n$ (ambient space).

Optimized Dynamic Mode Decomposition (**Askham et al.**)

Stack all the data together:

$$\mathbf{X} = [\mathbf{x}(t_1) \mid \mathbf{x}(t_2) \cdots \mid \mathbf{x}(t_p)] \in \mathbb{R}^{n \times p}$$

Optimize directly for DMD matrix decomposition:

$$\min_{\phi_j, \omega_j, b_j} \left\| \mathbf{X} - \sum_{j=1}^r \phi_j e^{\omega_j t} b_j \right\|_F$$

Translate to:

$$\min_{\mathbf{A}} \|\mathbf{X}^\top - \mathbf{\Phi}(\boldsymbol{\alpha})\mathbf{B}\|_F$$

Optimized Dynamic Mode Decomposition (continued)

Equivalent elementwise model:

$$\mathbf{x}_{i,:}^\top \approx \sum_{j=1}^r e^{\alpha_j t_i} \mathbf{B}_{j,:}$$

- $\Phi(\alpha)_{i,j} = \exp(\alpha_j t_i)$
- $b_j = \|\mathbf{B}^\top(:, j)\|_2$
- $\phi_j = \frac{\mathbf{B}^\top(:, j)}{b_j}$

Solve via variable projection method.

Further Reading

There are MANY variants of DMD

- Multiverse of DMD
- Physics-informed DMD
- Generalizing DMD: Modern Koopman theory

Tutorial

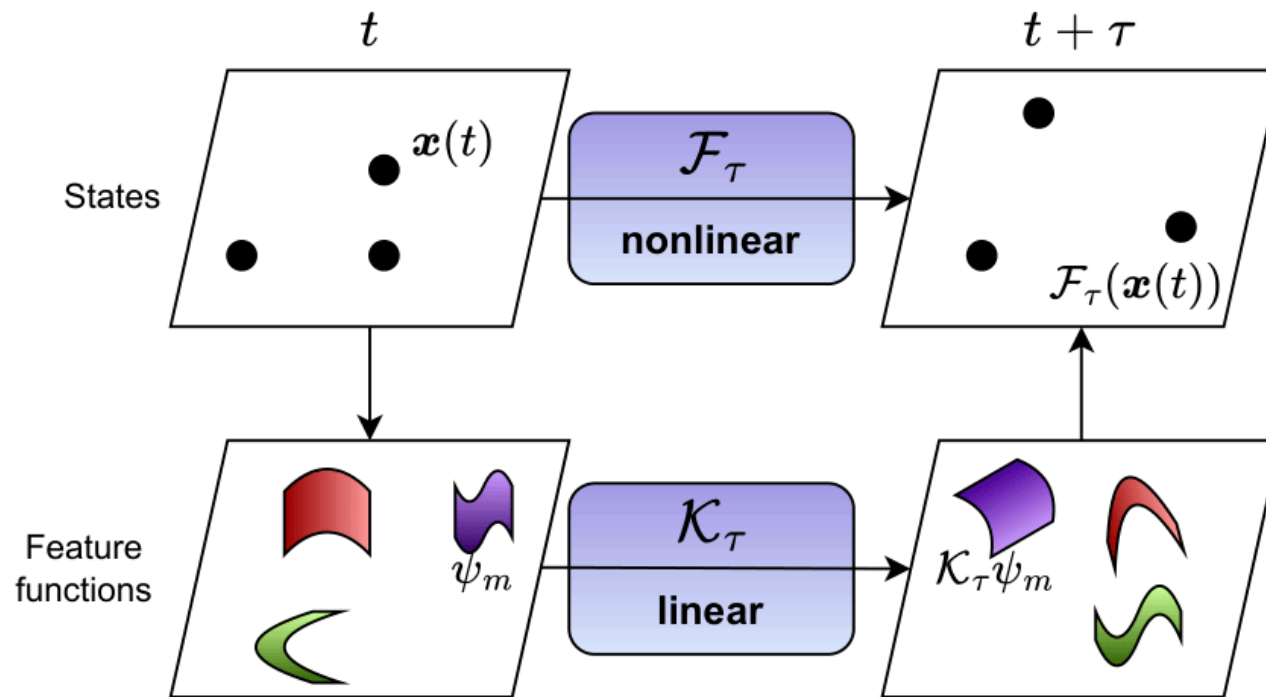
<https://github.com/PyDMD/PyDMD/blob/master/tutorials/tutorial1/tutorial-1-dmd.ipynb>

Koopman Mode Decomposition

Koopman Mode Decomposition

Problem: there's no linear map that goes from $\mathbf{x}(t)$ to $\mathbf{x}(t + \tau)$...

Solution:



Koopman Mode Decomposition

ψ_m “observable” (a.k.a. feature) function

Koopman operator (denoted $\mathcal{K}_\tau$) ([Koopman 1931](#)):

$$\mathcal{K}_\tau[\psi_m](\mathbf{x}(t)) = \psi_m(\mathcal{F}_\tau \mathbf{x}(t)) = \psi_m(\mathbf{x}(t + \tau))$$

Koopman Mode Decomposition (continued 2)

Koopman mode decomposition ([Rowley et al. 2009](#), [Mezić 2013](#)):

$$\mathbf{x}(t) = \sum_{j=1}^r \mathbf{b}_j e^{\omega_j t} \phi_j(\mathbf{x}(0))$$

- $\mathbf{b}_j$ **spatial patterns** (dynamic modes)
- ω_j **temporal characteristics** (continuous time eigenvalues)
- ϕ_j Koopman eigenfunctions

Koopman Mode Decomposition



Koopman Mode Decomposition (continued)

Koopman operator (denoted $\mathcal{K}_\tau$) ([Koopman 1931](#)):

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Koopman mode decomposition ([Rowley et al. 2009](#), [Mezić 2013](#)):

$$\mathbf{x}(t) = \sum_{j=1}^r \mathbf{b}_j e^{\omega_j t} \phi_j(\mathbf{x}(0))$$

- $\mathbf{b}_j$ **spatial patterns** (dynamic modes)
- ω_j **temporal characteristics** (continuous time eigenvalues)
- ϕ_j **scaling** (Koopman eigenfunctions)

Eigenfunction relation used implicitly:

$$\mathcal{K}_\tau[\phi_j] = e^{\omega_j \tau} \phi_j$$

Kernel Koopman Mode Decomposition (Williams et al. 2015, Klus et al. 2017)

- Embed states into a Reproducing Kernel Hilbert Space (RKHS), denoted $\mathcal{H}$
- Let $\psi : \mathbb{R}^M \rightarrow \mathcal{H}$ be the feature map for some kernel $k(\cdot, \cdot)$
- Define the snapshot feature matrices:

$$\begin{aligned}\Psi_{\mathbf{X}} &= [\psi(\mathbf{x}(1)) \cdots \psi(\mathbf{x}(T - \tau))] \\ \Psi_{\mathbf{X}'} &= [\psi(\mathbf{x}(\tau + 1)) \cdots \psi(\mathbf{x}'(T))]\end{aligned}$$

Kernel Koopman Mode Decomposition (continued)

- Seek finite-dimensional Koopman operator estimate $\tilde{\mathbf{K}}$, found via kernel Gram matrices:

$$\mathbf{K}_{\mathbf{X}} = \Psi_{\mathbf{X}}^{\top} \Psi_{\mathbf{X}}, \quad \mathbf{K}_{\mathbf{X}', \mathbf{X}} = \Psi_{\mathbf{X}'}^{\top} \Psi_{\mathbf{X}}$$

Kernel Koopman Mode Decomposition

- Using the eigendecomposition $\mathbf{K}_X = \mathbf{Q}\Sigma^2\mathbf{Q}^\top$
- The finite-dimensional Koopman operator estimate is

$$\tilde{\mathbf{K}} = \Sigma^{-1}\mathbf{Q}^\top \mathbf{K}_{X',X} \mathbf{Q}\Sigma^{-1}$$

- Eigenvalue decomposition $\tilde{\mathbf{K}}\mathbf{W} = \mathbf{W}\Lambda$

Kernel Koopman Mode Decomposition (continued)

Name	Definition
Eigenfunctions of $\mathbf{X}$	$\Phi = \mathbf{K}_X \mathbf{Q} \Sigma^\dagger \mathbf{W}$
Koopman modes	$\mathbf{B} = \mathbf{X} \mathbf{Q}^\top \Sigma^\dagger \mathbf{W}^{-1}$

Tutorial

- Extended (Kernel) DMD
- Koopman Mode Decomposition